

ABHIJITH SIVAPRASADAN

Energy Systems & Software Engineer | Heat Pump Intelligence | Python / Backend Workflows

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PROFILE

Energy systems and software engineer completing an M.Sc. in Sustainable Energy Engineering at KTH, with prior backend engineering experience and strong interest in Weheat Intelligence, installer/homeowner digital products and technical product support. Hybrid profile spanning APIs, testing, Python data workflows, heat-pump and building-energy modelling, operational energy data, validation mindset and clear technical communication. Seeking to contribute to Weheat's Utrecht founding team across application software, technical product/support workflows and the bridge between customers, installers and Eindhoven engineering teams.

CORE FIT FOR WEHEAT UTRECHT

- Application software and product workflows: backend APIs, automated tests, Git, Postman, debugging, documentation and maintainable delivery from professional software engineering experience.
 - Technical product/support bridge: translate heat-pump behaviour, telemetry, anomalies and validation findings into clear user value for installers, homeowners and internal teams.
 - Heat-pump intelligence context: building heat demand, heating-system modelling, energy optimisation, KPI thinking, data quality and CO2/cost-performance framing.
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EXPERIENCE

Backend Engineer

QBurst, Thiruvananthapuram, India

Aug 2021 - Jul 2023

- Developed and maintained backend APIs, data workflows and automated test pipelines using Python, TypeScript, Postman, Git and CI-oriented development practices.
- Translated requirements and change requests into maintainable software deliverables with focus on debugging, reproducibility, version control and test discipline.
- Built transferable experience for application software and installer/consumer digital products: API thinking, validation workflows, technical documentation and reliable delivery.

Energy Efficiency Intern

Alleima AB, Sandviken (Remote)

Dec 2024 - Jun 2025

- Built ISO 50001/EED-aligned energy performance mapping with KPI/EnPI definitions, baselines, load drivers and normalisation logic for comparing operating states.
- Assessed metering and data readiness; proposed measurement plans for reliable energy follow-up, anomaly detection and improvement tracking.
- Translated noisy operational energy data into structured findings and improvement roadmaps, strengthening ability to support data-informed product/service workflows.

Test Engineer / Master Thesis Student

Siemens Energy AB, Fluid Dynamic Lab, Finspång

Apr 2025 - Nov 2025

- Developed CFD/CHT models for high-temperature test-rig hardware, defining KPIs, assumptions and validation limitations.
- Commissioned NI-DAQ measurement chains with thermocouples and pressure sensors; refined LabVIEW acquisition workflows and linked simulation outputs to measurable quantities.
- Documented assumptions, validation limits and verification needs - transferable to product feedback loops, technical support and engineering-to-customer communication.

Student Intern - Pyrolysis Research

KTH Royal Institute of Technology, Stockholm

Feb 2024 - Jun 2024

- Conducted cost-analysis research and reviewed plastic-pyrolysis reactor concepts to support technology assessment and early-stage process evaluation.
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RELEVANT PROJECTS

Day-Ahead Building Heating Demand Forecasting

KTH Course Project (Python, scikit-learn, time-series)

2025

- Built heating-demand forecasting workflows using weather and temporal features; compared baseline and neural-network models with feature engineering, error analysis and model evaluation.
- Relevant to Weheat: turning heat and home-context data into predictions, diagnostics and explainable insight for digital products and support workflows.

Heating Dispatch Optimisation - District Heating System

KTH Course Project (Python, PuLP, MILP)

2025

- Implemented multi-asset heat-supply dispatch optimisation with operational constraints, heat-pump/CHP operation, electricity-price coupling, CO2 cost, storage and KPI reporting.
- Analysed cost/emissions trade-offs and flexibility value, building intuition for user-facing energy savings, CO2 reduction and grid-aware heating logic.

Smart Energy Island and Residential Heating Modelling

KTH Course Projects (IDA ICE, HOMER Pro, SAM)

2024 - 2025

- Modelled heat demand, PV, batteries, heat pumps, solar thermal, TES and district heating; evaluated CAPEX, OPEX, LCOE, payback, autonomy and system performance.

EDUCATION

M.Sc. in Sustainable Energy Engineering (expected 2026)

KTH Royal Institute of Technology, Stockholm

2023 - 2026

Relevant coursework: Energy in the Built Environment, Energy Management, AI Applications in Sustainable Energy Engineering, Practical Optimization of Energy Networks, Modelling of Energy Systems, Numerical Heat Transfer and Applied Heat and Power Technology.

B.Tech in Mechanical Engineering

APJ Abdul Kalam Technological University, Kerala, India

2017 - 2021

Relevant foundation: thermodynamics, heat and mass transfer, fluid mechanics, thermal engineering, mechanical systems and CAD/CAE.

TECHNICAL SKILLS

Software / backend:	Python, TypeScript, REST APIs, Postman, Git, Docker, automated tests, debugging, reproducible workflows, API validation and maintainable delivery.
Data / analytics:	pandas, NumPy, SciPy, scikit-learn, matplotlib, time-series workflows, feature engineering, error analysis, data quality and anomaly/loss-mechanism analysis.
Energy / heat pumps:	Heat-pump and heating-system fundamentals, building heat-demand modelling, COP/SCOP reasoning, KPI/EnPI frameworks, energy savings and CO2/cost framing.
Product/support:	Technical documentation, translating complex engineering findings into clear recommendations, installer/homeowner empathy, support-process thinking and stakeholder communication.
Optimisation/tools:	Python/PuLP MILP, operational constraints, cost/CO2 objectives, electricity-price coupling, IDA ICE, HOMER Pro, SAM, MATLAB basics, ANSYS Fluent, NI-DAQ/LabVIEW.

ADDITIONAL INFORMATION

Certifications: SSG Entre / Industrial Safety | Applied CFD - Siemens | IBM Python for Data Science | Digital Manufacturing & Design Technology | Schneider Electric sustainability course

Leadership: Founder of SAEINDIA Collegiate Club chapter; SAE team lead/captain and secretary/treasurer roles; Students' Nobel NightCap operations and safety role.

Languages: English - fluent/professional | Malayalam - native | Swedish - beginner, actively studying